

City of Detroit 2022 - 2025 Sales

Elliot Dean - Program Analyst IV
Office of the Chief Financial Officer | Office of the Assessor
City of Detroit

23 July 2026

Contents

Introduction	1
Model I	1
Model II	2
Model III	3
Model IV	10

Introduction

This report evaluates the metrics of a linear regression model created from citywide datasets of sales data from the years 2022 - 2024 and 2023 - 2025. Overlapping sales IDs were removed to include only unique sales, leaving a dataset containing single family residential sales from 2022 - 2025. After removing outliers in the initial data prep, a total of 19756 sales were used for model development.

Custom functions were written to automate variable creation from initial datasets, calculate assessing coefficients (PRD, COD, PRB), variance factors, and graphs.

```
## <environment: 0x000002c66eac1968>
```

Functions Used: modfunctions_notes.R + modelfunctions.R

R Data File: cod_2022-2025.RData Original Data Files: salesSFApr2023toMarch2025Updated.csv + SFResiSalesJUN2022-MAY2024_w-Exterior1.csv

Model I

Table 1: LN_SPPSF model fit statistics

n	df_resid	df_reg	df_total	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
19756	19732	23	19755	24	0.53508	0.1621	0.16112	0.53476	0.4318	165.9671	0	31383	31580

Table 2: LN_SPPSF model coefficients (95% CI)

term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	3.9561	221.1662	0.0179	0.0000	3.9211	3.9912
single_family1story	0.0509	4.8473	0.0105	0.0000	0.0303	0.0715
half_duplex1story	0.0816	1.7381	0.0470	0.0822	-0.0104	0.1737
smallest	0.1295	8.6521	0.0150	0.0000	0.1002	0.1588
small	0.0530	4.9302	0.0107	0.0000	0.0319	0.0740
two_plus_baths	0.1798	7.9969	0.0225	0.0000	0.1357	0.2238

term	estimate	statistic	std_error	p_value	conf_low	conf_high
powder_room	0.0773	6.1889	0.0125	0.0000	0.0528	0.1018
good	0.3570	12.1469	0.0294	0.0000	0.2994	0.4146
fair	-0.2064	-21.7091	0.0095	0.0000	-0.2250	-0.1877
poor	-0.3558	-16.3789	0.0217	0.0000	-0.3984	-0.3132
very_poor_unsound	-0.3324	-7.3201	0.0454	0.0000	-0.4214	-0.2434
qabove_avg	0.4421	15.5956	0.0284	0.0000	0.3866	0.4977
qbelow_avg	-0.0337	-3.4557	0.0098	0.0005	-0.0528	-0.0146
yb_pre_1929	-0.1747	-15.3768	0.0114	0.0000	-0.1970	-0.1525
yb_1946_to_1960	0.0758	8.1864	0.0093	0.0000	0.0577	0.0940
ln_garagespaces	0.1034	10.4659	0.0099	0.0000	0.0840	0.1227
fireplace	0.0984	10.5351	0.0093	0.0000	0.0801	0.1167
hot_water	0.0727	4.8905	0.0149	0.0000	0.0436	0.1019
elec_baseboard	-0.3213	-2.2362	0.1437	0.0253	-0.6029	-0.0397
wall	-0.1214	-2.5250	0.0481	0.0116	-0.2156	-0.0272
central_air	0.0927	8.6230	0.0107	0.0000	0.0716	0.1137
months_1to9	-0.0102	-4.8981	0.0021	0.0000	-0.0143	-0.0061
months_10to16	0.0187	8.5256	0.0022	0.0000	0.0144	0.0230
months_17to24	0.0099	10.4231	0.0010	0.0000	0.0081	0.0118

Model II

Table 3: LN_PRICE model fit statistics

n	df_resid	df_reg	df_total	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic
19756	19729	26	19755	27	0.53281	0.37118	0.37035	0.53245	0.42952	447.9058	0	31218 31

Table 4: LN_PRICE model coefficients (95% CI)

term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	10.9441	566.5569	0.0193	0.0000	10.9062	10.9820
ln_landratio	0.2301	13.0188	0.0177	0.0000	0.1955	0.2648
ln_base_bldgsize_ratio	0.9017	22.3717	0.0403	0.0000	0.8227	0.9807
single_family1story	0.0409	3.7183	0.0110	0.0002	0.0193	0.0624
half_duplex1story	-0.1416	-3.0049	0.0471	0.0027	-0.2340	-0.0492
half_duplex2story	-0.1799	-3.9578	0.0454	0.0001	-0.2690	-0.0908
smallest	0.0863	3.8109	0.0226	0.0001	0.0419	0.1307
small	0.0313	2.2824	0.0137	0.0225	0.0044	0.0582
two_plus_baths	0.1701	5.6599	0.0300	0.0000	0.1112	0.2290
powder_room	0.0850	5.3503	0.0159	0.0000	0.0539	0.1162
good	0.3603	12.3082	0.0293	0.0000	0.3029	0.4176
fair	-0.2029	-21.4219	0.0095	0.0000	-0.2215	-0.1844
poor	-0.3507	-16.2074	0.0216	0.0000	-0.3931	-0.3083
very_poor_unsound	-0.3272	-7.2352	0.0452	0.0000	-0.4158	-0.2385
qabove_avg	0.4123	14.4135	0.0286	0.0000	0.3562	0.4683
qbelow_avg	-0.0387	-3.8977	0.0099	0.0001	-0.0581	-0.0192
yb_pre_1929	-0.1449	-12.4982	0.0116	0.0000	-0.1676	-0.1221
yb_1946_to_1960	0.0593	6.3699	0.0093	0.0000	0.0411	0.0776
ln_garagespaces	0.0815	8.1511	0.0100	0.0000	0.0619	0.1011
fireplace	0.0959	10.1820	0.0094	0.0000	0.0775	0.1144
hot_water	0.0715	4.8188	0.0148	0.0000	0.0424	0.1006
elec_baseboard	-0.3596	-2.5127	0.1431	0.0120	-0.6401	-0.0791

term	estimate	statistic	std_error	p_value	conf_low	conf_high
wall	-0.1338	-2.7932	0.0479	0.0052	-0.2277	-0.0399
central_air	0.0910	8.4893	0.0107	0.0000	0.0700	0.1120
months_1to9	-0.0101	-4.8473	0.0021	0.0000	-0.0141	-0.0060
months_10to16	0.0189	8.6121	0.0022	0.0000	0.0146	0.0232
months_17to24	0.0101	10.6913	0.0009	0.0000	0.0083	0.0120

Model III

Table 5: LN_PRICE model (base) assessing coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	within_10_pct	within_20_pct	within_50_pct
19756	0.9849147	1.115474	0.9058778	1.231374	41.59922	-0.5587016	15.08403	30.71472	73.52197

Table 6: LN_PRICE (base) model fit statistics

n	df_resid	df_reg	df_total	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic
19756	19569	186	19755	187	0.47111	0.51237	0.50774	0.46888	0.38115	110.5488	0	26514

Table 7: LN_PRICE (+ECFs) model coefficients (95% CI)

term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	11.3167	519.7985	0.0218	0.0000	11.2740	11.3594
ln_landratio	0.1120	6.2150	0.0180	0.0000	0.0767	0.1474
ln_base_bldgsize_ratio	0.6862	18.4451	0.0372	0.0000	0.6133	0.7591
single_family1story	0.0313	3.1756	0.0099	0.0015	0.0120	0.0507
half_duplex1story	-0.1649	-3.8605	0.0427	0.0001	-0.2486	-0.0812
half_duplex2story	-0.2345	-5.7106	0.0411	0.0000	-0.3150	-0.1540
smallest	0.0444	2.1644	0.0205	0.0304	0.0042	0.0846
small	0.0242	1.9470	0.0124	0.0515	-0.0002	0.0485
two_plus_baths	0.0914	3.2925	0.0278	0.0010	0.0370	0.1458
powder_room	0.0282	1.9551	0.0144	0.0506	-0.0001	0.0566
good	0.2952	11.2329	0.0263	0.0000	0.2437	0.3467
fair	-0.1628	-19.0295	0.0086	0.0000	-0.1796	-0.1460
poor	-0.2918	-15.0666	0.0194	0.0000	-0.3297	-0.2538
very_poor_unsound	-0.2331	-5.7616	0.0405	0.0000	-0.3124	-0.1538
qabove_avg	0.1898	6.3880	0.0297	0.0000	0.1316	0.2481
qbelow_avg	-0.0588	-6.0011	0.0098	0.0000	-0.0780	-0.0396
yb_pre_1929	-0.0741	-6.1807	0.0120	0.0000	-0.0976	-0.0506
yb_1946_to_1960	0.0450	5.0395	0.0089	0.0000	0.0275	0.0625
yb_post_1961	0.0732	2.9722	0.0246	0.0030	0.0249	0.1215
crawl	-0.0798	-3.3483	0.0238	0.0008	-0.1265	-0.0331
slab	-0.1075	-3.8377	0.0280	0.0001	-0.1624	-0.0526
ln_garagespaces	0.1020	11.2718	0.0091	0.0000	0.0843	0.1198
fireplace	0.0626	6.8385	0.0092	0.0000	0.0447	0.0806
hot_water	0.0423	3.1448	0.0134	0.0017	0.0159	0.0686
wall	-0.1707	-3.7726	0.0453	0.0002	-0.2595	-0.0820
central_air	0.0483	5.0327	0.0096	0.0000	0.0295	0.0672
ecf_1R102	-0.3751	-5.6840	0.0660	0.0000	-0.5044	-0.2457
ecf_1R104	-0.1180	-3.8799	0.0304	0.0001	-0.1776	-0.0584
ecf_1R105	-0.3179	-9.2124	0.0345	0.0000	-0.3855	-0.2502

term	estimate	statistic	std_error	p_value	conf_low	conf_high
ecf_1R106	-0.2759	-5.1164	0.0539	0.0000	-0.3817	-0.1702
ecf_1R107	-0.2204	-2.2664	0.0972	0.0234	-0.4110	-0.0298
ecf_1R108	-0.4366	-12.4087	0.0352	0.0000	-0.5056	-0.3676
ecf_1R109	-0.1453	-4.9353	0.0294	0.0000	-0.2030	-0.0876
ecf_1R110	-0.1329	-2.8018	0.0474	0.0051	-0.2258	-0.0399
ecf_1R112	-0.1447	-2.5975	0.0557	0.0094	-0.2539	-0.0355
ecf_1R113	-0.5516	-9.2824	0.0594	0.0000	-0.6681	-0.4351
ecf_1R114	-0.3163	-5.0703	0.0624	0.0000	-0.4386	-0.1941
ecf_1R116	0.1654	4.0654	0.0407	0.0000	0.0857	0.2452
ecf_1R117	-0.1380	-4.0161	0.0344	0.0001	-0.2054	-0.0707
ecf_1R118	-0.2481	-6.0990	0.0407	0.0000	-0.3279	-0.1684
ecf_1R119	-0.3842	-8.4506	0.0455	0.0000	-0.4733	-0.2951
ecf_1R120	-0.1099	-1.8309	0.0600	0.0671	-0.2275	0.0078
ecf_1R121	-0.1947	-5.2589	0.0370	0.0000	-0.2673	-0.1222
ecf_1R122	-0.3847	-9.5320	0.0404	0.0000	-0.4638	-0.3056
ecf_1R123	-0.3310	-6.2254	0.0532	0.0000	-0.4352	-0.2267
ecf_1R124	-0.7050	-9.4355	0.0747	0.0000	-0.8514	-0.5585
ecf_1R126	0.2500	5.6198	0.0445	0.0000	0.1628	0.3372
ecf_1R127	-0.3658	-6.1888	0.0591	0.0000	-0.4816	-0.2499
ecf_1R128	-0.4858	-12.7109	0.0382	0.0000	-0.5607	-0.4109
ecf_1R129	-0.1754	-3.4724	0.0505	0.0005	-0.2745	-0.0764
ecf_1R130	-0.5997	-3.3161	0.1809	0.0009	-0.9542	-0.2452
ecf_1R131	-0.3392	-3.4791	0.0975	0.0005	-0.5303	-0.1481
ecf_1R133	-0.3911	-4.8998	0.0798	0.0000	-0.5475	-0.2346
ecf_1R135	-0.2394	-5.4750	0.0437	0.0000	-0.3252	-0.1537
ecf_1R136	-0.2877	-7.2732	0.0396	0.0000	-0.3652	-0.2102
ecf_1R137	-0.7252	-12.2223	0.0593	0.0000	-0.8415	-0.6089
ecf_2R201	-0.2940	-10.3037	0.0285	0.0000	-0.3499	-0.2380
ecf_2R202	-0.1099	-2.1127	0.0520	0.0346	-0.2119	-0.0079
ecf_2R205	-0.2016	-6.1677	0.0327	0.0000	-0.2656	-0.1375
ecf_2R208	0.4455	6.4056	0.0695	0.0000	0.3092	0.5818
ecf_2R209	0.4115	3.8298	0.1074	0.0001	0.2009	0.6220
ecf_2R210	0.4725	3.3230	0.1422	0.0009	0.1938	0.7511
ecf_2R211	-1.1341	-5.3604	0.2116	0.0000	-1.5487	-0.7194
ecf_2R212	-0.1823	-6.1135	0.0298	0.0000	-0.2408	-0.1239
ecf_2R213	-0.0888	-2.6747	0.0332	0.0075	-0.1540	-0.0237
ecf_2R214	0.0787	3.6075	0.0218	0.0003	0.0359	0.1215
ecf_2R215	0.2982	5.0191	0.0594	0.0000	0.1818	0.4147
ecf_2R218	-0.5561	-7.2378	0.0768	0.0000	-0.7067	-0.4055
ecf_2R219	0.2939	2.2321	0.1317	0.0256	0.0358	0.5519
ecf_2R220	-0.2543	-6.8613	0.0371	0.0000	-0.3269	-0.1816
ecf_2R221	-0.1789	-5.4248	0.0330	0.0000	-0.2435	-0.1143
ecf_2R222	-0.3658	-10.9284	0.0335	0.0000	-0.4314	-0.3002
ecf_2R223	-0.2669	-7.2310	0.0369	0.0000	-0.3393	-0.1946
ecf_2R224	-0.4723	-12.0687	0.0391	0.0000	-0.5491	-0.3956
ecf_2R225	-0.2852	-7.5470	0.0378	0.0000	-0.3593	-0.2112
ecf_2R226	-0.6501	-5.9545	0.1092	0.0000	-0.8641	-0.4361
ecf_2R228	-0.8333	-6.5596	0.1270	0.0000	-1.0823	-0.5843
ecf_2R229	-0.6446	-6.3398	0.1017	0.0000	-0.8439	-0.4453
ecf_2R230	-0.7903	-4.0904	0.1932	0.0000	-1.1691	-0.4116
ecf_3R301	-0.6902	-8.2685	0.0835	0.0000	-0.8538	-0.5266
ecf_3R302	-0.2823	-1.9733	0.1431	0.0485	-0.5628	-0.0019
ecf_3R303	-0.5966	-18.7380	0.0318	0.0000	-0.6590	-0.5342
ecf_3R304	-0.8435	-13.9468	0.0605	0.0000	-0.9621	-0.7250
ecf_3R305	-0.8727	-7.3268	0.1191	0.0000	-1.1061	-0.6392
ecf_3R307	-0.4622	-14.1768	0.0326	0.0000	-0.5261	-0.3983
ecf_3R308	-0.7535	-10.5121	0.0717	0.0000	-0.8940	-0.6130

term	estimate	statistic	std_error	p_value	conf_low	conf_high
ecf_3R309	-0.7655	-10.8343	0.0707	0.0000	-0.9040	-0.6270
ecf_3R310	0.2269	5.2790	0.0430	0.0000	0.1427	0.3112
ecf_3R311	-0.5132	-7.1435	0.0718	0.0000	-0.6540	-0.3724
ecf_3R312	-0.7227	-17.5835	0.0411	0.0000	-0.8033	-0.6422
ecf_3R313	-0.6663	-14.3024	0.0466	0.0000	-0.7576	-0.5750
ecf_3R314	-0.3977	-7.0886	0.0561	0.0000	-0.5077	-0.2877
ecf_3R315	-0.3735	-8.1666	0.0457	0.0000	-0.4631	-0.2838
ecf_3R316	-0.6843	-11.8029	0.0580	0.0000	-0.7980	-0.5707
ecf_3R317	-1.0100	-7.6796	0.1315	0.0000	-1.2678	-0.7522
ecf_3R318	-0.3075	-2.3371	0.1316	0.0194	-0.5654	-0.0496
ecf_3R319	-0.5957	-21.2537	0.0280	0.0000	-0.6506	-0.5408
ecf_3R320	-0.5139	-14.5424	0.0353	0.0000	-0.5832	-0.4447
ecf_3R321	-0.8679	-14.7764	0.0587	0.0000	-0.9831	-0.7528
ecf_3R322	-0.2796	-5.9828	0.0467	0.0000	-0.3712	-0.1880
ecf_3R323	-0.4574	-13.7331	0.0333	0.0000	-0.5226	-0.3921
ecf_3R324	-0.7409	-10.8814	0.0681	0.0000	-0.8743	-0.6074
ecf_3R325	-0.5472	-11.4424	0.0478	0.0000	-0.6409	-0.4535
ecf_3R326	-1.1466	-14.5050	0.0791	0.0000	-1.3016	-0.9917
ecf_4R401	-0.6256	-16.1403	0.0388	0.0000	-0.7016	-0.5497
ecf_4R402	-0.9857	-16.3746	0.0602	0.0000	-1.1036	-0.8677
ecf_4R403	-0.5003	-11.4472	0.0437	0.0000	-0.5860	-0.4146
ecf_4R404	-0.4830	-10.9900	0.0439	0.0000	-0.5691	-0.3968
ecf_4R405	-0.3891	-12.8737	0.0302	0.0000	-0.4484	-0.3299
ecf_4R406	-0.4406	-15.2762	0.0288	0.0000	-0.4971	-0.3841
ecf_4R407	-0.6092	-11.0721	0.0550	0.0000	-0.7171	-0.5014
ecf_4R408	-0.5711	-16.8034	0.0340	0.0000	-0.6377	-0.5045
ecf_4R409	-0.4492	-8.7657	0.0512	0.0000	-0.5497	-0.3488
ecf_4R410	-0.1772	-6.6600	0.0266	0.0000	-0.2294	-0.1251
ecf_4R411	-0.6598	-19.8227	0.0333	0.0000	-0.7250	-0.5945
ecf_4R412	-0.7612	-13.8959	0.0548	0.0000	-0.8686	-0.6538
ecf_4R413	-0.2910	-8.5381	0.0341	0.0000	-0.3578	-0.2242
ecf_4R415	0.1744	3.2948	0.0529	0.0010	0.0707	0.2782
ecf_4R416	-0.7553	-7.9273	0.0953	0.0000	-0.9421	-0.5686
ecf_4R417	-0.1517	-4.5376	0.0334	0.0000	-0.2172	-0.0862
ecf_4R418	-0.4641	-4.1327	0.1123	0.0000	-0.6842	-0.2440
ecf_4R419	-0.6959	-12.1967	0.0571	0.0000	-0.8077	-0.5841
ecf_4R421	-0.7224	-10.0138	0.0721	0.0000	-0.8638	-0.5810
ecf_4R422	0.0975	2.1433	0.0455	0.0321	0.0083	0.1867
ecf_5R501	-0.6763	-14.4706	0.0467	0.0000	-0.7679	-0.5847
ecf_5R502	-0.5335	-12.5925	0.0424	0.0000	-0.6166	-0.4505
ecf_5R503	0.4407	4.2660	0.1033	0.0000	0.2382	0.6432
ecf_5R504	0.3421	5.2954	0.0646	0.0000	0.2155	0.4687
ecf_5R507	-0.4328	-4.5754	0.0946	0.0000	-0.6182	-0.2474
ecf_5R508	0.5356	2.7400	0.1955	0.0061	0.1524	0.9187
ecf_5R509	0.3988	3.9560	0.1008	0.0001	0.2012	0.5964
ecf_5R510	-0.3523	-4.1347	0.0852	0.0000	-0.5193	-0.1853
ecf_5R514	0.2275	3.7721	0.0603	0.0002	0.1093	0.3457
ecf_5R517	0.6276	3.7423	0.1677	0.0002	0.2989	0.9564
ecf_5R519	0.7836	2.3470	0.3339	0.0189	0.1292	1.4380
ecf_5R521	-0.2255	-2.5312	0.0891	0.0114	-0.4002	-0.0509
ecf_5R523	-0.5805	-6.3790	0.0910	0.0000	-0.7589	-0.4021
ecf_5R524	1.1467	3.3997	0.3373	0.0007	0.4856	1.8078
ecf_5R528	-0.6080	-8.5122	0.0714	0.0000	-0.7479	-0.4680
ecf_5R529	0.2097	3.6064	0.0582	0.0003	0.0957	0.3237
ecf_5R530	0.6909	4.8123	0.1436	0.0000	0.4095	0.9722
ecf_5R531	-0.1578	-2.1906	0.0720	0.0285	-0.2990	-0.0166
ecf_5R532	0.4090	3.3427	0.1224	0.0008	0.1692	0.6488

term	estimate	statistic	std_error	p_value	conf_low	conf_high
ecf_5R533	-0.7943	-10.8572	0.0732	0.0000	-0.9377	-0.6509
ecf_5R534	-0.4291	-6.0253	0.0712	0.0000	-0.5687	-0.2895
ecf_5R535	0.5845	2.9980	0.1949	0.0027	0.2023	0.9666
ecf_5R538	-0.6493	-5.1136	0.1270	0.0000	-0.8982	-0.4004
ecf_6R601	-1.0482	-13.5852	0.0772	0.0000	-1.1994	-0.8969
ecf_6R602	-0.8167	-15.2937	0.0534	0.0000	-0.9214	-0.7121
ecf_6R603	-0.9105	-15.6640	0.0581	0.0000	-1.0244	-0.7965
ecf_6R604	-0.4842	-10.2751	0.0471	0.0000	-0.5765	-0.3918
ecf_6R605	-0.2841	-4.5830	0.0620	0.0000	-0.4057	-0.1626
ecf_6R606	-0.4470	-5.1017	0.0876	0.0000	-0.6188	-0.2753
ecf_6R608	0.7742	5.8328	0.1327	0.0000	0.5141	1.0344
ecf_6R609	1.1179	4.7235	0.2367	0.0000	0.6540	1.5818
ecf_6R611	-0.1431	-3.2692	0.0438	0.0011	-0.2290	-0.0573
ecf_6R612	-0.2718	-3.2907	0.0826	0.0010	-0.4337	-0.1099
ecf_6R614	1.1380	8.2562	0.1378	0.0000	0.8678	1.4081
ecf_6R616	0.5062	2.8291	0.1789	0.0047	0.1555	0.8568
ecf_6R617	0.3533	1.6666	0.2120	0.0956	-0.0622	0.7688
ecf_6R618	-0.5333	-4.6159	0.1155	0.0000	-0.7598	-0.3068
ecf_6R620	-0.5242	-15.6656	0.0335	0.0000	-0.5898	-0.4586
ecf_7R701	0.1308	2.7454	0.0476	0.0060	0.0374	0.2241
ecf_7R703	-0.5508	-9.3403	0.0590	0.0000	-0.6664	-0.4352
ecf_7R704	-0.3985	-14.5623	0.0274	0.0000	-0.4521	-0.3449
ecf_7R705	-0.4020	-19.4344	0.0207	0.0000	-0.4425	-0.3614
ecf_7R706	-0.3741	-5.9278	0.0631	0.0000	-0.4977	-0.2504
ecf_7R707	-0.4921	-7.2249	0.0681	0.0000	-0.6256	-0.3586
ecf_7R708	-0.4847	-13.2508	0.0366	0.0000	-0.5564	-0.4130
ecf_7R709	-0.4606	-12.2226	0.0377	0.0000	-0.5344	-0.3867
ecf_7R710	-0.2451	-6.2539	0.0392	0.0000	-0.3219	-0.1683
ecf_7R711	-0.5300	-15.4082	0.0344	0.0000	-0.5974	-0.4626
ecf_7R712	-0.4812	-8.5491	0.0563	0.0000	-0.5916	-0.3709
ecf_7R713	-0.4296	-8.9496	0.0480	0.0000	-0.5237	-0.3355
ecf_7R714	-0.8620	-4.8199	0.1788	0.0000	-1.2125	-0.5115
ecf_7R715	-0.6680	-7.6539	0.0873	0.0000	-0.8390	-0.4969
ecf_7R716	-0.9989	-18.2508	0.0547	0.0000	-1.1062	-0.8916
ecf_7R717	-0.9433	-19.2305	0.0491	0.0000	-1.0395	-0.8472
ecf_7R718	-0.6854	-17.9229	0.0382	0.0000	-0.7603	-0.6104
ecf_7R719	-0.3199	-11.9340	0.0268	0.0000	-0.3725	-0.2674
ecf_7R720	-0.7949	-7.4558	0.1066	0.0000	-1.0039	-0.5859
ecf_7R722	-0.6913	-7.7729	0.0889	0.0000	-0.8657	-0.5170
ecf_7R723	-0.1770	-3.3190	0.0533	0.0009	-0.2815	-0.0725
ecf_7R724	-0.7457	-11.5151	0.0648	0.0000	-0.8726	-0.6188
months_1to9	-0.0087	-4.7216	0.0018	0.0000	-0.0123	-0.0051
months_10to16	0.0166	8.5161	0.0019	0.0000	0.0128	0.0204
months_17to24	0.0110	13.0720	0.0008	0.0000	0.0094	0.0127

Table 8: VIF / GVIF

term	vif	tolerance
ln_base_bldgsizes_ratio	10.432	0.096
smallest	3.353	0.298
months_10to16	3.274	0.305
two_plus_baths	3.056	0.327
small	2.577	0.388
yb_pre_1929	2.446	0.409
powder_room	2.355	0.425

term	vif	tolerance
qbelow_avg	2.094	0.478
qabove_avg	2.052	0.487
months_1to9	2.024	0.494
months_17to24	2.017	0.496
ecf_7R705	1.783	0.561
fireplace	1.756	0.570
ln_landratio	1.740	0.575
ecf_2R214	1.703	0.587
single_family1story	1.700	0.588
yb_1946_to_1960	1.624	0.616
hot_water	1.537	0.650
ecf_7R704	1.326	0.754
ecf_4R410	1.321	0.757
ecf_3R319	1.312	0.762
ecf_2R201	1.310	0.763
ecf_2R215	1.299	0.770
ecf_7R719	1.294	0.773
ecf_4R406	1.281	0.780
ecf_1R104	1.267	0.789
ecf_2R212	1.261	0.793
yb_post_1961	1.251	0.799
ecf_4R405	1.248	0.801
ecf_1R109	1.245	0.803
ecf_2R205	1.239	0.807
crawl	1.222	0.818
ecf_2R221	1.219	0.820
ecf_5R504	1.218	0.821
ecf_3R303	1.216	0.822
ecf_2R222	1.212	0.825
ecf_3R307	1.206	0.829
ecf_3R323	1.204	0.830
ecf_1R105	1.204	0.831
ecf_2R213	1.203	0.831
ecf_6R620	1.196	0.836
ecf_1R117	1.194	0.837
ecf_4R417	1.194	0.838
ecf_4R408	1.193	0.838
ecf_7R711	1.186	0.843
ecf_4R413	1.185	0.844
ln_garagespaces	1.179	0.848
ecf_6R611	1.175	0.851
ecf_3R320	1.175	0.851
ecf_4R411	1.174	0.852
ecf_2R225	1.167	0.857
wall	1.165	0.859
ecf_1R108	1.164	0.859
ecf_7R708	1.164	0.859
ecf_2R220	1.158	0.863
ecf_3R310	1.157	0.864
ecf_2R224	1.156	0.865
ecf_5R502	1.156	0.865
ecf_2R223	1.155	0.866
ecf_7R709	1.154	0.867
ecf_4R404	1.149	0.870
ecf_1R121	1.144	0.874
half_duplex1story	1.143	0.875
ecf_1R128	1.142	0.876

term	vif	tolerance
ecf_1R116	1.140	0.877
ecf_7R710	1.139	0.878
half_duplex2story	1.138	0.879
ecf_4R403	1.137	0.880
ecf_7R718	1.136	0.880
ecf_5R501	1.135	0.881
ecf_6R604	1.134	0.882
ecf_4R401	1.128	0.887
ecf_3R312	1.126	0.888
ecf_1R118	1.125	0.889
central_air	1.121	0.892
slab	1.120	0.893
ecf_1R136	1.119	0.894
ecf_1R122	1.114	0.898
ecf_7R717	1.111	0.900
ecf_1R110	1.108	0.902
ecf_3R325	1.107	0.904
ecf_5R503	1.105	0.905
ecf_1R135	1.104	0.905
ecf_4R422	1.103	0.906
fair	1.102	0.908
ecf_4R421	1.100	0.909
ecf_1R126	1.099	0.910
ecf_3R313	1.099	0.910
ecf_7R701	1.098	0.911
ecf_7R723	1.097	0.912
ecf_3R322	1.096	0.912
ecf_3R315	1.096	0.913
ecf_7R713	1.095	0.914
ecf_5R529	1.093	0.915
ecf_1R119	1.092	0.915
ecf_2R210	1.092	0.916
ecf_2R209	1.091	0.917
ecf_6R602	1.087	0.920
ecf_1R129	1.087	0.920
ecf_1R112	1.086	0.921
ecf_4R419	1.081	0.925
ecf_5R532	1.078	0.927
ecf_5R514	1.078	0.928
ecf_1R123	1.078	0.928
ecf_6R603	1.077	0.929
ecf_7R716	1.075	0.930
ecf_4R409	1.072	0.933
ecf_6R605	1.070	0.934
ecf_1R106	1.070	0.934
ecf_4R415	1.069	0.936
ecf_2R202	1.068	0.936
ecf_3R304	1.068	0.936
ecf_7R712	1.067	0.937
ecf_4R412	1.064	0.940
ecf_1R113	1.062	0.941
poor	1.062	0.941
ecf_7R703	1.062	0.942
ecf_4R407	1.060	0.944
ecf_3R314	1.060	0.944
ecf_1R137	1.059	0.944
ecf_5R533	1.059	0.945

term	vif	tolerance
ecf_4R402	1.058	0.945
ecf_1R102	1.057	0.946
ecf_3R316	1.056	0.947
ecf_7R706	1.055	0.948
ecf_7R724	1.055	0.948
ecf_5R528	1.055	0.948
ecf_3R309	1.055	0.948
good	1.054	0.948
ecf_1R124	1.054	0.949
ecf_3R321	1.053	0.949
ecf_5R509	1.052	0.951
ecf_1R120	1.051	0.951
ecf_1R127	1.051	0.951
ecf_5R531	1.050	0.953
ecf_1R114	1.049	0.953
ecf_5R534	1.049	0.953
ecf_5R507	1.047	0.955
ecf_5R510	1.045	0.957
ecf_3R311	1.044	0.958
ecf_6R601	1.044	0.958
ecf_6R612	1.043	0.958
ecf_2R208	1.043	0.958
ecf_5R523	1.043	0.959
ecf_7R707	1.043	0.959
ecf_3R324	1.042	0.960
very_poor_unsound	1.040	0.962
ecf_3R326	1.040	0.962
ecf_3R308	1.039	0.962
ecf_6R606	1.036	0.965
ecf_5R521	1.036	0.965
ecf_2R218	1.035	0.966
ecf_3R301	1.034	0.967
ecf_5R508	1.033	0.969
ecf_7R722	1.032	0.969
ecf_1R133	1.031	0.970
ecf_6R608	1.031	0.970
ecf_1R130	1.031	0.970
ecf_7R715	1.028	0.973
ecf_5R535	1.027	0.974
ecf_1R131	1.027	0.974
ecf_6R614	1.027	0.974
ecf_5R524	1.025	0.976
ecf_2R229	1.024	0.977
ecf_7R720	1.023	0.977
ecf_3R305	1.022	0.979
ecf_4R418	1.022	0.979
ecf_6R618	1.022	0.979
ecf_4R416	1.021	0.979
ecf_1R107	1.021	0.979
ecf_5R530	1.021	0.980
ecf_2R226	1.019	0.981
ecf_2R228	1.017	0.983
ecf_5R538	1.016	0.984
ecf_2R219	1.015	0.986
ecf_3R302	1.014	0.986
ecf_5R517	1.013	0.987
ecf_3R318	1.013	0.987

term	vif	tolerance
ecf_3R317	1.013	0.988
ecf_6R617	1.012	0.988
ecf_6R616	1.009	0.991
ecf_6R609	1.009	0.991
ecf_2R230	1.009	0.991
ecf_7R714	1.008	0.992
ecf_2R211	1.008	0.992
ecf_5R519	1.004	0.996

Model IV

Table 9: LN_TASP model assessing coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	within_10_pct	within_20_pct	within_50_pct
18400	1.163368	1.286896	1.092881	1.177527	37.00051	-0.5791432	16.83152	33.23913	68.88043

Table 10: Alternative LN_TASP model assessing coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB	within_10_pct	within_20_pct	within_50_pct
18399	1.165305	1.286947	1.092218	1.178288	36.94861	-0.5796658	16.76178	33.24637	68.79722

Table 11: LN_TASP model fit statistics

n	df_resid	df_reg	df_total	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic	bic
18400	18215	184	18399	187	0.40735	0.56776	0.5634	0.4053	0.33911	130.0333	0	19353	20848

Table 12: LN_TASP model coefficients (95% CI)

term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	11.5695	611.1977	0.0189	0.0000	11.5324	11.6066
ln_landratio	0.1244	7.4313	0.0167	0.0000	0.0916	0.1572
ln_base_bldgsize_ratio	0.7235	21.0273	0.0344	0.0000	0.6561	0.7910
single_family1story	0.0203	2.3301	0.0087	0.0198	0.0032	0.0374
half_duplex1story	-0.2147	-5.5867	0.0384	0.0000	-0.2901	-0.1394
half_duplex2story	-0.2980	-8.0784	0.0369	0.0000	-0.3703	-0.2257
smallest	0.0433	2.3287	0.0186	0.0199	0.0069	0.0797
small	0.0336	3.0119	0.0111	0.0026	0.0117	0.0554
two_plus_baths	0.0790	3.0633	0.0258	0.0022	0.0284	0.1295
powder_room	0.0230	1.7357	0.0133	0.0826	-0.0030	0.0490
good	0.2576	10.4201	0.0247	0.0000	0.2092	0.3061
fair	-0.1522	-19.7254	0.0077	0.0000	-0.1673	-0.1371
poor	-0.2872	-15.4638	0.0186	0.0000	-0.3236	-0.2508
very_poor_unsound	-0.2260	-5.2663	0.0429	0.0000	-0.3101	-0.1419
qabove_avg	0.1497	5.4363	0.0275	0.0000	0.0957	0.2036
qbelow_avg	-0.0595	-6.7708	0.0088	0.0000	-0.0767	-0.0423
yb_pre_1929	-0.0733	-6.7345	0.0109	0.0000	-0.0946	-0.0519
yb_1946_to_1960	0.0419	5.3167	0.0079	0.0000	0.0264	0.0573
yb_post_1961	0.0917	4.0786	0.0225	0.0000	0.0476	0.1358

term	estimate	statistic	std_error	p_value	conf_low	conf_high
crawl	-0.0864	-3.7598	0.0230	0.0002	-0.1315	-0.0414
slab	-0.1320	-4.9438	0.0267	0.0000	-0.1844	-0.0797
ln_garagespaces	0.0980	12.0027	0.0082	0.0000	0.0820	0.1140
fireplace	0.0633	7.7005	0.0082	0.0000	0.0472	0.0795
hot_water	0.0400	3.2035	0.0125	0.0014	0.0155	0.0644
wall	-0.2201	-5.0484	0.0436	0.0000	-0.3055	-0.1346
central_air	0.0506	5.9874	0.0084	0.0000	0.0340	0.0671
ecf_1R102	-0.3288	-5.3037	0.0620	0.0000	-0.4503	-0.2073
ecf_1R104	-0.1329	-4.8273	0.0275	0.0000	-0.1868	-0.0789
ecf_1R105	-0.3344	-10.7846	0.0310	0.0000	-0.3952	-0.2736
ecf_1R106	-0.2726	-5.6323	0.0484	0.0000	-0.3674	-0.1777
ecf_1R107	-0.2519	-2.5930	0.0971	0.0095	-0.4423	-0.0615
ecf_1R108	-0.4637	-14.5526	0.0319	0.0000	-0.5262	-0.4013
ecf_1R109	-0.1616	-6.0346	0.0268	0.0000	-0.2141	-0.1091
ecf_1R110	-0.1379	-3.2502	0.0424	0.0012	-0.2210	-0.0547
ecf_1R112	-0.1373	-2.7818	0.0494	0.0054	-0.2341	-0.0406
ecf_1R113	-0.5585	-9.7237	0.0574	0.0000	-0.6711	-0.4459
ecf_1R114	-0.2967	-5.0897	0.0583	0.0000	-0.4110	-0.1825
ecf_1R116	0.1294	3.5883	0.0361	0.0003	0.0587	0.2001
ecf_1R117	-0.1620	-5.2469	0.0309	0.0000	-0.2225	-0.1015
ecf_1R118	-0.2475	-6.7659	0.0366	0.0000	-0.3192	-0.1758
ecf_1R119	-0.4671	-11.4546	0.0408	0.0000	-0.5470	-0.3871
ecf_1R120	-0.1154	-2.1342	0.0541	0.0328	-0.2214	-0.0094
ecf_1R121	-0.2078	-6.2478	0.0333	0.0000	-0.2730	-0.1426
ecf_1R122	-0.4056	-11.1957	0.0362	0.0000	-0.4766	-0.3346
ecf_1R123	-0.3039	-6.2643	0.0485	0.0000	-0.3990	-0.2088
ecf_1R124	-0.6375	-8.6050	0.0741	0.0000	-0.7827	-0.4923
ecf_1R125	-0.1191	-1.8873	0.0631	0.0591	-0.2428	0.0046
ecf_1R126	0.2244	5.6885	0.0394	0.0000	0.1471	0.3017
ecf_1R127	-0.4047	-7.6007	0.0532	0.0000	-0.5091	-0.3003
ecf_1R128	-0.5356	-15.5146	0.0345	0.0000	-0.6032	-0.4679
ecf_1R129	-0.1841	-4.1419	0.0444	0.0000	-0.2712	-0.0970
ecf_1R130	NA	NA	NA	NA	NA	NA
ecf_1R131	-0.3796	-3.5627	0.1066	0.0004	-0.5885	-0.1708
ecf_1R133	-0.4314	-5.3104	0.0812	0.0000	-0.5906	-0.2721
ecf_1R135	-0.2530	-6.4862	0.0390	0.0000	-0.3294	-0.1765
ecf_1R136	-0.3121	-8.8720	0.0352	0.0000	-0.3810	-0.2431
ecf_1R137	-0.7734	-13.2370	0.0584	0.0000	-0.8879	-0.6589
ecf_2R201	-0.3146	-12.0875	0.0260	0.0000	-0.3657	-0.2636
ecf_2R202	-0.1471	-3.1869	0.0462	0.0014	-0.2376	-0.0566
ecf_2R203	-0.0496	-1.7894	0.0277	0.0736	-0.1039	0.0047
ecf_2R205	-0.2133	-7.2043	0.0296	0.0000	-0.2714	-0.1553
ecf_2R208	0.4768	7.3749	0.0647	0.0000	0.3501	0.6035
ecf_2R209	0.3335	3.2337	0.1031	0.0012	0.1313	0.5356
ecf_2R210	0.3957	2.7990	0.1414	0.0051	0.1186	0.6729
ecf_2R211	-1.0890	-3.7656	0.2892	0.0002	-1.6558	-0.5221
ecf_2R212	-0.1971	-7.2756	0.0271	0.0000	-0.2501	-0.1440
ecf_2R213	-0.1070	-3.5710	0.0300	0.0004	-0.1657	-0.0483
ecf_2R214	0.0508	2.4505	0.0207	0.0143	0.0102	0.0915
ecf_2R215	0.3010	5.6738	0.0530	0.0000	0.1970	0.4049
ecf_2R216	0.2693	1.7056	0.1579	0.0881	-0.0402	0.5789
ecf_2R217	-0.4043	-1.7078	0.2368	0.0877	-0.8684	0.0597
ecf_2R218	-0.5972	-6.9257	0.0862	0.0000	-0.7662	-0.4282
ecf_2R219	0.2677	1.8457	0.1450	0.0650	-0.0166	0.5519
ecf_2R220	-0.2688	-8.0347	0.0335	0.0000	-0.3344	-0.2032
ecf_2R221	-0.2088	-6.9949	0.0299	0.0000	-0.2673	-0.1503
ecf_2R222	-0.3740	-12.3317	0.0303	0.0000	-0.4335	-0.3146

term	estimate	statistic	std_error	p_value	conf_low	conf_high
ecf_2R223	-0.2860	-8.6079	0.0332	0.0000	-0.3511	-0.2209
ecf_2R224	-0.4824	-13.6434	0.0354	0.0000	-0.5517	-0.4131
ecf_2R225	-0.3084	-8.8693	0.0348	0.0000	-0.3766	-0.2403
ecf_2R226	-0.6645	-5.1148	0.1299	0.0000	-0.9191	-0.4098
ecf_2R228	-0.9840	-6.3471	0.1550	0.0000	-1.2879	-0.6801
ecf_2R229	-0.7189	-5.2585	0.1367	0.0000	-0.9869	-0.4510
ecf_2R230	-0.9088	-2.2262	0.4082	0.0260	-1.7090	-0.1086
ecf_3R301	-0.5928	-4.9935	0.1187	0.0000	-0.8255	-0.3601
ecf_3R303	-0.6149	-21.0449	0.0292	0.0000	-0.6722	-0.5577
ecf_3R304	-0.8293	-12.9339	0.0641	0.0000	-0.9550	-0.7036
ecf_3R305	-1.0313	-6.1630	0.1673	0.0000	-1.3593	-0.7033
ecf_3R306	0.1667	3.1668	0.0526	0.0015	0.0635	0.2699
ecf_3R307	-0.4837	-16.4147	0.0295	0.0000	-0.5415	-0.4259
ecf_3R308	-0.8127	-10.8852	0.0747	0.0000	-0.9590	-0.6663
ecf_3R309	-0.8281	-10.6999	0.0774	0.0000	-0.9798	-0.6764
ecf_3R310	0.3539	8.7688	0.0404	0.0000	0.2748	0.4330
ecf_3R311	-0.5290	-7.0697	0.0748	0.0000	-0.6756	-0.3823
ecf_3R312	-0.7647	-20.6527	0.0370	0.0000	-0.8373	-0.6922
ecf_3R313	-0.7017	-16.6458	0.0422	0.0000	-0.7843	-0.6190
ecf_3R314	-0.3547	-6.6501	0.0533	0.0000	-0.4592	-0.2501
ecf_3R315	-0.3931	-9.6166	0.0409	0.0000	-0.4732	-0.3130
ecf_3R316	-0.6207	-10.3512	0.0600	0.0000	-0.7383	-0.5032
ecf_3R317	-0.9743	-5.3266	0.1829	0.0000	-1.3328	-0.6158
ecf_3R318	-0.3444	-2.9025	0.1187	0.0037	-0.5770	-0.1118
ecf_3R319	-0.6355	-24.7268	0.0257	0.0000	-0.6858	-0.5851
ecf_3R320	-0.5428	-16.9510	0.0320	0.0000	-0.6056	-0.4801
ecf_3R321	-0.8384	-14.6368	0.0573	0.0000	-0.9506	-0.7261
ecf_3R322	-0.2984	-7.2314	0.0413	0.0000	-0.3793	-0.2175
ecf_3R323	-0.4819	-16.0738	0.0300	0.0000	-0.5406	-0.4231
ecf_3R324	-0.7965	-11.9126	0.0669	0.0000	-0.9275	-0.6654
ecf_3R325	-0.5541	-12.9469	0.0428	0.0000	-0.6380	-0.4702
ecf_3R326	-1.2016	-15.7805	0.0761	0.0000	-1.3508	-1.0523
ecf_4R401	-0.6636	-18.9930	0.0349	0.0000	-0.7321	-0.5951
ecf_4R402	-1.0661	-18.2282	0.0585	0.0000	-1.1808	-0.9515
ecf_4R403	-0.5422	-13.8053	0.0393	0.0000	-0.6192	-0.4652
ecf_4R404	-0.4931	-12.5574	0.0393	0.0000	-0.5701	-0.4162
ecf_4R405	-0.3992	-14.5170	0.0275	0.0000	-0.4531	-0.3453
ecf_4R406	-0.4688	-17.7467	0.0264	0.0000	-0.5206	-0.4170
ecf_4R407	-0.5862	-11.4073	0.0514	0.0000	-0.6869	-0.4855
ecf_4R408	-0.5919	-19.2643	0.0307	0.0000	-0.6521	-0.5316
ecf_4R409	-0.4826	-10.6512	0.0453	0.0000	-0.5714	-0.3938
ecf_4R410	-0.1931	-7.8448	0.0246	0.0000	-0.2414	-0.1449
ecf_4R411	-0.6614	-21.5205	0.0307	0.0000	-0.7217	-0.6012
ecf_4R412	-0.7557	-14.1582	0.0534	0.0000	-0.8604	-0.6511
ecf_4R413	-0.3131	-10.1916	0.0307	0.0000	-0.3733	-0.2529
ecf_4R415	0.1721	3.6633	0.0470	0.0002	0.0800	0.2641
ecf_4R416	-0.6861	-6.4508	0.1064	0.0000	-0.8945	-0.4776
ecf_4R417	-0.1699	-5.6110	0.0303	0.0000	-0.2293	-0.1106
ecf_4R418	-0.5873	-4.5190	0.1300	0.0000	-0.8420	-0.3326
ecf_4R419	-0.8183	-14.3795	0.0569	0.0000	-0.9299	-0.7068
ecf_4R421	-0.7425	-10.8819	0.0682	0.0000	-0.8763	-0.6088
ecf_4R422	0.1140	2.7461	0.0415	0.0060	0.0326	0.1955
ecf_5R501	-0.7884	-17.1372	0.0460	0.0000	-0.8786	-0.6982
ecf_5R502	-0.6737	-16.0949	0.0419	0.0000	-0.7557	-0.5916
ecf_5R503	0.4793	4.8774	0.0983	0.0000	0.2867	0.6719
ecf_5R504	0.3806	6.5653	0.0580	0.0000	0.2669	0.4942
ecf_5R505	0.4146	2.0189	0.2054	0.0435	0.0121	0.8171

term	estimate	statistic	std_error	p_value	conf_low	conf_high
ecf_5R507	-0.5009	-4.5274	0.1106	0.0000	-0.7177	-0.2840
ecf_5R508	0.6806	1.6642	0.4090	0.0961	-0.1210	1.4822
ecf_5R509	0.4167	3.7460	0.1112	0.0002	0.1987	0.6347
ecf_5R510	-0.7541	-5.5055	0.1370	0.0000	-1.0226	-0.4857
ecf_5R514	0.2118	3.4659	0.0611	0.0005	0.0920	0.3316
ecf_5R517	0.5891	3.2136	0.1833	0.0013	0.2298	0.9484
ecf_5R519	0.7442	2.5771	0.2888	0.0100	0.1782	1.3102
ecf_5R521	-0.3768	-2.5960	0.1451	0.0094	-0.6612	-0.0923
ecf_5R523	-0.6378	-4.3859	0.1454	0.0000	-0.9228	-0.3527
ecf_5R524	NA	NA	NA	NA	NA	NA
ecf_5R528	-0.6947	-9.3953	0.0739	0.0000	-0.8396	-0.5498
ecf_5R529	0.2641	4.3829	0.0603	0.0000	0.1460	0.3822
ecf_5R530	0.6571	2.2724	0.2892	0.0231	0.0903	1.2239
ecf_5R531	-0.2411	-2.5424	0.0948	0.0110	-0.4270	-0.0552
ecf_5R532	0.4623	3.3052	0.1399	0.0010	0.1881	0.7364
ecf_5R533	-0.8555	-10.2591	0.0834	0.0000	-1.0190	-0.6921
ecf_5R534	-0.5298	-6.1226	0.0865	0.0000	-0.6994	-0.3602
ecf_5R538	-0.8110	-4.4331	0.1829	0.0000	-1.1695	-0.4524
ecf_5R539	-0.3083	-1.9884	0.1550	0.0468	-0.6122	-0.0044
ecf_6R601	-1.1281	-14.0908	0.0801	0.0000	-1.2851	-0.9712
ecf_6R602	-0.8790	-16.3368	0.0538	0.0000	-0.9844	-0.7735
ecf_6R603	-1.0497	-16.6693	0.0630	0.0000	-1.1731	-0.9263
ecf_6R604	-0.5165	-11.9417	0.0432	0.0000	-0.6012	-0.4317
ecf_6R605	-0.2864	-4.7268	0.0606	0.0000	-0.4051	-0.1676
ecf_6R606	-0.5069	-4.5985	0.1102	0.0000	-0.7230	-0.2908
ecf_6R608	0.8509	6.1692	0.1379	0.0000	0.5805	1.1212
ecf_6R609	1.0886	2.6695	0.4078	0.0076	0.2893	1.8880
ecf_6R611	-0.1566	-3.9593	0.0396	0.0001	-0.2342	-0.0791
ecf_6R612	-0.2823	-3.2585	0.0866	0.0011	-0.4520	-0.1125
ecf_6R614	1.0022	5.4497	0.1839	0.0000	0.6417	1.3626
ecf_6R618	-0.5894	-3.7994	0.1551	0.0001	-0.8935	-0.2853
ecf_6R620	-0.5306	-17.3943	0.0305	0.0000	-0.5904	-0.4708
ecf_7R701	0.1211	2.8725	0.0421	0.0041	0.0385	0.2037
ecf_7R703	-0.5899	-11.2728	0.0523	0.0000	-0.6925	-0.4874
ecf_7R704	-0.4207	-16.7949	0.0251	0.0000	-0.4698	-0.3716
ecf_7R705	-0.4102	-20.8399	0.0197	0.0000	-0.4488	-0.3716
ecf_7R706	-0.4549	-7.8524	0.0579	0.0000	-0.5685	-0.3414
ecf_7R707	-0.5029	-7.7003	0.0653	0.0000	-0.6309	-0.3749
ecf_7R708	-0.5165	-15.6487	0.0330	0.0000	-0.5812	-0.4518
ecf_7R709	-0.4706	-13.8882	0.0339	0.0000	-0.5371	-0.4042
ecf_7R710	-0.2684	-7.6464	0.0351	0.0000	-0.3372	-0.1996
ecf_7R711	-0.5455	-17.5922	0.0310	0.0000	-0.6063	-0.4847
ecf_7R712	-0.4489	-8.8584	0.0507	0.0000	-0.5483	-0.3496
ecf_7R713	-0.4568	-10.5983	0.0431	0.0000	-0.5413	-0.3723
ecf_7R714	-0.9384	-2.3008	0.4079	0.0214	-1.7378	-0.1389
ecf_7R715	-0.6383	-6.0056	0.1063	0.0000	-0.8467	-0.4300
ecf_7R716	-0.9558	-17.5479	0.0545	0.0000	-1.0625	-0.8490
ecf_7R717	-1.0384	-22.5990	0.0459	0.0000	-1.1285	-0.9484
ecf_7R718	-0.7148	-20.5209	0.0348	0.0000	-0.7831	-0.6465
ecf_7R719	-0.3382	-13.7008	0.0247	0.0000	-0.3866	-0.2898
ecf_7R720	-0.8272	-6.9672	0.1187	0.0000	-1.0599	-0.5945
ecf_7R722	-0.7761	-6.5326	0.1188	0.0000	-1.0089	-0.5432
ecf_7R723	-0.1395	-2.7877	0.0501	0.0053	-0.2376	-0.0414
ecf_7R724	-0.7557	-10.7080	0.0706	0.0000	-0.8941	-0.6174

Table 13: Alternative LN_TASP model fit statistics

n	df_resid	df_reg	df_total	p	see	r2	adj_r2	rmse	mae	f_stat	f_p	aic
18399	18212	186	18398	189	0.40754	0.56742	0.563	0.40547	0.33924	128.4361	0	19371

Table 14: Alternative LN_TASP model coefficients (95% CI)

term	estimate	statistic	std_error	p_value	conf_low	conf_high
(Intercept)	9.0726	51.7200	0.1754	0.0000	8.7288	9.4165
ln_landratio	0.1374	8.2553	0.0166	0.0000	0.1048	0.1701
lnsbldsfint	0.3872	14.5527	0.0266	0.0000	0.3351	0.4394
single_family1story	0.0234	2.7168	0.0086	0.0066	0.0065	0.0402
half_duplex2story	-0.1256	-3.4136	0.0368	0.0006	-0.1977	-0.0535
smallest	0.0554	2.3649	0.0234	0.0180	0.0095	0.1013
small	0.0247	2.0811	0.0119	0.0374	0.0014	0.0479
large	0.0313	2.5034	0.0125	0.0123	0.0068	0.0559
largest	0.0729	3.2343	0.0225	0.0012	0.0287	0.1171
two_plus_baths	0.1414	5.4258	0.0261	0.0000	0.0903	0.1925
powder_room	0.0371	2.7036	0.0137	0.0069	0.0102	0.0639
good	0.2599	10.5052	0.0247	0.0000	0.2114	0.3083
fair	-0.1517	-19.6448	0.0077	0.0000	-0.1668	-0.1366
poor	-0.2872	-15.4471	0.0186	0.0000	-0.3236	-0.2507
very_poor_unsound	-0.2289	-5.3316	0.0429	0.0000	-0.3131	-0.1448
qabove_avg	0.1468	5.2390	0.0280	0.0000	0.0919	0.2017
qbelow_avg	-0.0591	-6.7172	0.0088	0.0000	-0.0764	-0.0419
yb_pre_1929	-0.0692	-6.3500	0.0109	0.0000	-0.0905	-0.0478
yb_1946_to_1960	0.0441	5.5866	0.0079	0.0000	0.0286	0.0595
yb_post_1961	0.0950	4.2202	0.0225	0.0000	0.0509	0.1392
crawl	-0.0793	-3.4449	0.0230	0.0006	-0.1245	-0.0342
slab	-0.1270	-4.7551	0.0267	0.0000	-0.1794	-0.0747
garage	0.0789	11.3210	0.0070	0.0000	0.0652	0.0925
fireplace	0.0629	7.6437	0.0082	0.0000	0.0468	0.0790
hot_water	0.0372	2.9738	0.0125	0.0029	0.0127	0.0618
wall	-0.2155	-4.9375	0.0437	0.0000	-0.3011	-0.1300
central_air	0.0517	6.1258	0.0084	0.0000	0.0352	0.0683
ecf_1R102	-0.3332	-5.3723	0.0620	0.0000	-0.4547	-0.2116
ecf_1R104	-0.1302	-4.7275	0.0275	0.0000	-0.1842	-0.0762
ecf_1R105	-0.3341	-10.7681	0.0310	0.0000	-0.3949	-0.2733
ecf_1R106	-0.2732	-5.6416	0.0484	0.0000	-0.3681	-0.1783
ecf_1R107	-0.2502	-2.5745	0.0972	0.0100	-0.4407	-0.0597
ecf_1R108	-0.4611	-14.4558	0.0319	0.0000	-0.5236	-0.3985
ecf_1R109	-0.1604	-5.9849	0.0268	0.0000	-0.2130	-0.1079
ecf_1R110	-0.1410	-3.3232	0.0424	0.0009	-0.2242	-0.0578
ecf_1R112	-0.1321	-2.6734	0.0494	0.0075	-0.2289	-0.0352
ecf_1R113	-0.5559	-9.6723	0.0575	0.0000	-0.6685	-0.4432
ecf_1R114	-0.2957	-5.0680	0.0583	0.0000	-0.4100	-0.1813
ecf_1R116	0.1330	3.6805	0.0361	0.0002	0.0622	0.2039
ecf_1R117	-0.1615	-5.2228	0.0309	0.0000	-0.2221	-0.1009
ecf_1R118	-0.2477	-6.7662	0.0366	0.0000	-0.3194	-0.1759
ecf_1R119	-0.4664	-11.4275	0.0408	0.0000	-0.5464	-0.3864
ecf_1R120	-0.1213	-2.2407	0.0541	0.0251	-0.2273	-0.0152
ecf_1R121	-0.2070	-6.2175	0.0333	0.0000	-0.2723	-0.1418
ecf_1R122	-0.4036	-11.1296	0.0363	0.0000	-0.4746	-0.3325
ecf_1R123	-0.3006	-6.1934	0.0485	0.0000	-0.3958	-0.2055
ecf_1R124	-0.6356	-8.5744	0.0741	0.0000	-0.7809	-0.4903
ecf_1R125	-0.1234	-1.9550	0.0631	0.0506	-0.2471	0.0003

term	estimate	statistic	std_error	p_value	conf_low	conf_high
ecf_1R126	0.2214	5.6042	0.0395	0.0000	0.1439	0.2988
ecf_1R127	-0.4016	-7.5352	0.0533	0.0000	-0.5061	-0.2971
ecf_1R128	-0.5352	-15.4826	0.0346	0.0000	-0.6030	-0.4675
ecf_1R129	-0.1842	-4.1406	0.0445	0.0000	-0.2713	-0.0970
ecf_1R130	NA	NA	NA	NA	NA	NA
ecf_1R131	-0.3740	-3.5080	0.1066	0.0005	-0.5830	-0.1650
ecf_1R133	-0.4259	-5.2399	0.0813	0.0000	-0.5852	-0.2666
ecf_1R135	-0.2540	-6.5044	0.0390	0.0000	-0.3305	-0.1774
ecf_1R136	-0.3138	-8.9152	0.0352	0.0000	-0.3828	-0.2448
ecf_1R137	-0.7708	-13.1824	0.0585	0.0000	-0.8854	-0.6562
ecf_2R201	-0.3127	-12.0013	0.0261	0.0000	-0.3638	-0.2616
ecf_2R202	-0.1489	-3.2226	0.0462	0.0013	-0.2394	-0.0583
ecf_2R203	-0.0491	-1.7673	0.0278	0.0772	-0.1035	0.0054
ecf_2R205	-0.2098	-7.0779	0.0296	0.0000	-0.2679	-0.1517
ecf_2R208	0.4745	7.3308	0.0647	0.0000	0.3477	0.6014
ecf_2R209	0.3828	3.7015	0.1034	0.0002	0.1801	0.5855
ecf_2R210	0.6143	4.3424	0.1415	0.0000	0.3370	0.8917
ecf_2R211	-1.0821	-3.7397	0.2894	0.0002	-1.6493	-0.5149
ecf_2R212	-0.1985	-7.3191	0.0271	0.0000	-0.2516	-0.1453
ecf_2R213	-0.1063	-3.5454	0.0300	0.0004	-0.1651	-0.0475
ecf_2R214	0.0505	2.4201	0.0209	0.0155	0.0096	0.0915
ecf_2R215	0.3199	5.9723	0.0536	0.0000	0.2149	0.4248
ecf_2R216	0.3867	2.4450	0.1582	0.0145	0.0767	0.6967
ecf_2R218	-0.5901	-6.8380	0.0863	0.0000	-0.7593	-0.4210
ecf_2R219	0.2614	1.8009	0.1452	0.0717	-0.0231	0.5459
ecf_2R220	-0.2670	-7.9695	0.0335	0.0000	-0.3327	-0.2014
ecf_2R221	-0.2067	-6.9074	0.0299	0.0000	-0.2654	-0.1480
ecf_2R222	-0.3724	-12.2622	0.0304	0.0000	-0.4319	-0.3128
ecf_2R223	-0.2847	-8.5566	0.0333	0.0000	-0.3499	-0.2195
ecf_2R224	-0.4808	-13.5753	0.0354	0.0000	-0.5502	-0.4114
ecf_2R225	-0.3068	-8.8080	0.0348	0.0000	-0.3751	-0.2385
ecf_2R226	-0.6646	-5.1201	0.1298	0.0000	-0.9190	-0.4102
ecf_2R228	-0.9824	-6.3328	0.1551	0.0000	-1.2864	-0.6783
ecf_2R229	-0.7062	-5.1614	0.1368	0.0000	-0.9743	-0.4380
ecf_2R230	-0.9324	-2.2828	0.4085	0.0225	-1.7330	-0.1318
ecf_3R301	-0.5879	-4.9494	0.1188	0.0000	-0.8207	-0.3551
ecf_3R303	-0.6072	-20.7547	0.0293	0.0000	-0.6646	-0.5499
ecf_3R304	-0.8271	-12.8910	0.0642	0.0000	-0.9529	-0.7013
ecf_3R305	-1.0253	-6.1229	0.1674	0.0000	-1.3535	-0.6971
ecf_3R306	0.1694	3.2151	0.0527	0.0013	0.0661	0.2727
ecf_3R307	-0.4805	-16.2895	0.0295	0.0000	-0.5383	-0.4227
ecf_3R308	-0.8033	-10.7549	0.0747	0.0000	-0.9497	-0.6569
ecf_3R309	-0.8259	-10.6642	0.0774	0.0000	-0.9777	-0.6741
ecf_3R310	0.3565	8.7922	0.0405	0.0000	0.2770	0.4360
ecf_3R311	-0.5264	-7.0319	0.0749	0.0000	-0.6732	-0.3797
ecf_3R312	-0.7628	-20.5821	0.0371	0.0000	-0.8354	-0.6901
ecf_3R313	-0.6928	-16.4111	0.0422	0.0000	-0.7756	-0.6101
ecf_3R314	-0.3496	-6.5510	0.0534	0.0000	-0.4542	-0.2450
ecf_3R315	-0.3902	-9.5363	0.0409	0.0000	-0.4704	-0.3100
ecf_3R316	-0.6155	-10.2571	0.0600	0.0000	-0.7331	-0.4979
ecf_3R317	-0.9753	-5.3288	0.1830	0.0000	-1.3340	-0.6165
ecf_3R318	-0.3451	-2.9066	0.1187	0.0037	-0.5778	-0.1124
ecf_3R319	-0.6332	-24.6096	0.0257	0.0000	-0.6836	-0.5828
ecf_3R320	-0.5385	-16.8067	0.0320	0.0000	-0.6014	-0.4757
ecf_3R321	-0.8367	-14.5961	0.0573	0.0000	-0.9491	-0.7243
ecf_3R322	-0.2989	-7.2381	0.0413	0.0000	-0.3799	-0.2180
ecf_3R323	-0.4794	-15.9787	0.0300	0.0000	-0.5382	-0.4206

term	estimate	statistic	std_error	p_value	conf_low	conf_high
ecf_3R324	-0.7925	-11.8460	0.0669	0.0000	-0.9236	-0.6613
ecf_3R325	-0.5499	-12.8416	0.0428	0.0000	-0.6338	-0.4659
ecf_3R326	-1.2007	-15.7578	0.0762	0.0000	-1.3501	-1.0514
ecf_4R401	-0.6636	-18.9720	0.0350	0.0000	-0.7321	-0.5950
ecf_4R402	-1.0676	-18.2407	0.0585	0.0000	-1.1823	-0.9529
ecf_4R403	-0.5461	-13.9468	0.0392	0.0000	-0.6229	-0.4694
ecf_4R404	-0.4969	-12.7416	0.0390	0.0000	-0.5733	-0.4204
ecf_4R405	-0.4028	-14.6351	0.0275	0.0000	-0.4568	-0.3489
ecf_4R406	-0.4690	-17.7226	0.0265	0.0000	-0.5209	-0.4171
ecf_4R407	-0.5879	-11.4291	0.0514	0.0000	-0.6887	-0.4871
ecf_4R408	-0.5899	-19.1668	0.0308	0.0000	-0.6502	-0.5296
ecf_4R409	-0.4824	-10.6377	0.0453	0.0000	-0.5713	-0.3935
ecf_4R410	-0.1901	-7.7124	0.0247	0.0000	-0.2385	-0.1418
ecf_4R411	-0.6609	-21.4764	0.0308	0.0000	-0.7212	-0.6005
ecf_4R412	-0.7549	-14.1392	0.0534	0.0000	-0.8596	-0.6503
ecf_4R413	-0.3132	-10.1797	0.0308	0.0000	-0.3735	-0.2529
ecf_4R415	0.1725	3.6668	0.0470	0.0002	0.0803	0.2647
ecf_4R416	-0.6863	-6.4490	0.1064	0.0000	-0.8948	-0.4777
ecf_4R417	-0.1707	-5.6264	0.0303	0.0000	-0.2302	-0.1113
ecf_4R418	-0.5902	-4.5390	0.1300	0.0000	-0.8451	-0.3353
ecf_4R419	-0.8244	-14.4724	0.0570	0.0000	-0.9361	-0.7128
ecf_4R421	-0.7492	-10.9658	0.0683	0.0000	-0.8831	-0.6153
ecf_4R422	0.1160	2.7887	0.0416	0.0053	0.0345	0.1975
ecf_5R501	-0.7827	-16.9838	0.0461	0.0000	-0.8730	-0.6923
ecf_5R502	-0.6703	-15.9902	0.0419	0.0000	-0.7525	-0.5881
ecf_5R503	0.5580	5.6554	0.0987	0.0000	0.3646	0.7514
ecf_5R504	0.3887	6.6568	0.0584	0.0000	0.2742	0.5031
ecf_5R505	0.4061	1.9754	0.2056	0.0482	0.0032	0.8090
ecf_5R506	0.2034	1.6706	0.1217	0.0948	-0.0352	0.4419
ecf_5R507	-0.4984	-4.4994	0.1108	0.0000	-0.7155	-0.2813
ecf_5R508	0.8153	1.9918	0.4093	0.0464	0.0130	1.6176
ecf_5R509	0.4432	3.9778	0.1114	0.0001	0.2248	0.6617
ecf_5R510	-0.7531	-5.4962	0.1370	0.0000	-1.0217	-0.4845
ecf_5R514	0.2184	3.5653	0.0613	0.0004	0.0983	0.3385
ecf_5R517	0.5983	3.2621	0.1834	0.0011	0.2388	0.9578
ecf_5R519	0.7318	2.5326	0.2889	0.0113	0.1654	1.2981
ecf_5R521	-0.3721	-2.5618	0.1452	0.0104	-0.6568	-0.0874
ecf_5R523	-0.6379	-4.3843	0.1455	0.0000	-0.9231	-0.3527
ecf_5R524	NA	NA	NA	NA	NA	NA
ecf_5R528	-0.6886	-9.3050	0.0740	0.0000	-0.8336	-0.5435
ecf_5R529	0.2722	4.5113	0.0603	0.0000	0.1539	0.3905
ecf_5R530	0.6516	2.2521	0.2893	0.0243	0.0845	1.2186
ecf_5R531	-0.2331	-2.4560	0.0949	0.0141	-0.4192	-0.0471
ecf_5R532	0.6088	4.3437	0.1402	0.0000	0.3341	0.8835
ecf_5R533	-0.8603	-10.3049	0.0835	0.0000	-1.0239	-0.6966
ecf_5R534	-0.5333	-6.1580	0.0866	0.0000	-0.7030	-0.3635
ecf_5R535	0.5101	1.7579	0.2902	0.0788	-0.0587	1.0788
ecf_5R538	-0.8215	-4.4873	0.1831	0.0000	-1.1803	-0.4626
ecf_5R539	-0.3163	-2.0391	0.1551	0.0415	-0.6204	-0.0123
ecf_6R601	-1.1293	-14.0967	0.0801	0.0000	-1.2864	-0.9723
ecf_6R602	-0.8752	-16.2538	0.0538	0.0000	-0.9807	-0.7697
ecf_6R603	-1.0477	-16.6234	0.0630	0.0000	-1.1713	-0.9242
ecf_6R604	-0.5050	-11.6632	0.0433	0.0000	-0.5899	-0.4201
ecf_6R605	-0.2751	-4.5377	0.0606	0.0000	-0.3940	-0.1563
ecf_6R606	-0.4946	-4.4846	0.1103	0.0000	-0.7108	-0.2784
ecf_6R608	0.8560	6.2018	0.1380	0.0000	0.5855	1.1265
ecf_6R609	1.0634	2.6057	0.4081	0.0092	0.2635	1.8633

term	estimate	statistic	std_error	p_value	conf_low	conf_high
ecf_6R611	-0.1525	-3.8490	0.0396	0.0001	-0.2302	-0.0748
ecf_6R612	-0.2582	-2.9780	0.0867	0.0029	-0.4281	-0.0882
ecf_6R614	1.0075	5.4757	0.1840	0.0000	0.6469	1.3682
ecf_6R618	-0.5744	-3.7004	0.1552	0.0002	-0.8786	-0.2701
ecf_6R620	-0.5231	-17.1340	0.0305	0.0000	-0.5829	-0.4632
ecf_7R701	0.1208	2.8628	0.0422	0.0042	0.0381	0.2034
ecf_7R703	-0.5876	-11.2228	0.0524	0.0000	-0.6902	-0.4850
ecf_7R704	-0.4216	-16.8141	0.0251	0.0000	-0.4707	-0.3724
ecf_7R705	-0.4054	-20.5740	0.0197	0.0000	-0.4440	-0.3668
ecf_7R706	-0.4555	-7.8575	0.0580	0.0000	-0.5692	-0.3419
ecf_7R707	-0.4951	-7.5760	0.0653	0.0000	-0.6232	-0.3670
ecf_7R708	-0.5149	-15.5841	0.0330	0.0000	-0.5796	-0.4501
ecf_7R709	-0.4696	-13.8481	0.0339	0.0000	-0.5361	-0.4032
ecf_7R710	-0.2655	-7.5581	0.0351	0.0000	-0.3343	-0.1966
ecf_7R711	-0.5437	-17.5143	0.0310	0.0000	-0.6046	-0.4829
ecf_7R712	-0.4485	-8.8438	0.0507	0.0000	-0.5479	-0.3491
ecf_7R713	-0.4551	-10.5499	0.0431	0.0000	-0.5396	-0.3705
ecf_7R714	-0.9276	-2.2729	0.4081	0.0230	-1.7276	-0.1277
ecf_7R715	-0.6373	-5.9928	0.1063	0.0000	-0.8457	-0.4289
ecf_7R716	-0.9558	-17.5284	0.0545	0.0000	-1.0627	-0.8489
ecf_7R717	-1.0432	-22.6696	0.0460	0.0000	-1.1334	-0.9530
ecf_7R718	-0.7120	-20.4154	0.0349	0.0000	-0.7803	-0.6436
ecf_7R719	-0.3372	-13.6452	0.0247	0.0000	-0.3857	-0.2888
ecf_7R720	-0.8255	-6.9471	0.1188	0.0000	-1.0584	-0.5926
ecf_7R722	-0.7744	-6.5146	0.1189	0.0000	-1.0073	-0.5414
ecf_7R723	-0.1473	-2.9287	0.0503	0.0034	-0.2458	-0.0487
ecf_7R724	-0.7524	-10.6532	0.0706	0.0000	-0.8909	-0.6140

Table 15: VIF / GVIF

term	vif	tolerance
ln_base_bldgsize_ratio	10.606	0.094
smallest	3.411	0.293
two_plus_baths	3.016	0.332
small	2.619	0.382
powder_room	2.399	0.417
yb_pre_1929	2.330	0.429
ecf_7R705	2.114	0.473
qbelow_avg	2.104	0.475
qabove_avg	2.057	0.486
ecf_2R214	2.042	0.490
fireplace	1.775	0.563
ln_landratio	1.749	0.572
single_family1story	1.624	0.616
yb_1946_to_1960	1.603	0.624
hot_water	1.533	0.652
ecf_7R704	1.473	0.679
ecf_4R410	1.468	0.681
ecf_2R201	1.453	0.688
ecf_7R719	1.451	0.689
ecf_3R319	1.451	0.689
ecf_4R406	1.416	0.706
ecf_2R212	1.386	0.722
ecf_1R104	1.382	0.723
ecf_1R109	1.371	0.729

term	vif	tolerance
ecf_2R215	1.367	0.731
ecf_4R405	1.358	0.736
ecf_2R203	1.344	0.744
ecf_2R205	1.334	0.750
ecf_2R221	1.324	0.755
ecf_2R222	1.313	0.761
ecf_3R303	1.309	0.764
ecf_3R307	1.306	0.766
ecf_2R213	1.303	0.768
ecf_3R323	1.299	0.770
ecf_1R117	1.288	0.776
ecf_1R105	1.288	0.776
ecf_4R408	1.286	0.777
ecf_4R417	1.282	0.780
ecf_6R620	1.279	0.782
ecf_7R711	1.277	0.783
ecf_4R413	1.275	0.784
ecf_3R320	1.253	0.798
ecf_5R504	1.251	0.799
ecf_4R411	1.249	0.801
yb_post_1961	1.242	0.805
ecf_6R611	1.236	0.809
ecf_2R220	1.235	0.809
ecf_1R108	1.234	0.810
ecf_7R708	1.234	0.810
ecf_2R223	1.231	0.812
crawl	1.231	0.813
ecf_2R225	1.227	0.815
ecf_2R224	1.225	0.817
ecf_7R709	1.220	0.820
ecf_1R121	1.215	0.823
ecf_1R128	1.210	0.827
ecf_4R404	1.200	0.834
ecf_3R310	1.199	0.834
ecf_7R710	1.199	0.834
ecf_1R116	1.197	0.835
ecf_7R718	1.196	0.836
ecf_4R401	1.189	0.841
ecf_1R118	1.184	0.845
ecf_1R136	1.183	0.846
ecf_4R403	1.182	0.846
ecf_3R312	1.181	0.847
ecf_6R604	1.177	0.850
ecf_5R502	1.175	0.851
ecf_1R122	1.169	0.856
ln_garagespaces	1.169	0.856
ecf_1R135	1.156	0.865
wall	1.150	0.869
ecf_1R126	1.145	0.873
ecf_1R110	1.143	0.875
ecf_5R501	1.142	0.875
ecf_3R322	1.142	0.875
ecf_3R325	1.142	0.876
ecf_3R315	1.141	0.877
ecf_7R717	1.140	0.878
ecf_3R313	1.139	0.878
ecf_7R701	1.139	0.878

term	vif	tolerance
ecf_4R422	1.137	0.880
ecf_7R713	1.136	0.881
ecf_1R119	1.135	0.881
half_duplex1story	1.132	0.884
ecf_3R306	1.132	0.884
half_duplex2story	1.131	0.884
ecf_1R129	1.125	0.889
ecf_4R421	1.120	0.893
central_air	1.116	0.896
ecf_7R723	1.113	0.899
ecf_1R112	1.112	0.900
ecf_4R409	1.108	0.902
ecf_5R503	1.104	0.905
ecf_1R123	1.102	0.908
slab	1.101	0.908
ecf_2R202	1.099	0.910
ecf_4R415	1.099	0.910
ecf_1R106	1.096	0.912
ecf_6R602	1.095	0.913
ecf_7R712	1.095	0.913
ecf_5R529	1.091	0.917
ecf_4R419	1.090	0.918
ecf_2R209	1.089	0.919
fair	1.087	0.920
ecf_7R716	1.087	0.920
ecf_7R703	1.085	0.921
ecf_2R210	1.084	0.923
ecf_6R605	1.081	0.925
ecf_4R407	1.078	0.928
ecf_4R412	1.078	0.928
ecf_5R514	1.078	0.928
ecf_3R314	1.076	0.929
ecf_6R603	1.073	0.932
ecf_1R127	1.073	0.932
ecf_1R120	1.071	0.934
ecf_1R113	1.070	0.934
ecf_7R706	1.069	0.935
ecf_4R402	1.069	0.935
ecf_1R137	1.067	0.937
ecf_3R321	1.065	0.939
ecf_3R304	1.063	0.941
ecf_1R102	1.063	0.941
ecf_1R114	1.062	0.942
ecf_5R532	1.060	0.943
ecf_3R316	1.059	0.944
ecf_1R124	1.057	0.946
ecf_2R208	1.056	0.947
ecf_1R125	1.054	0.949
ecf_5R528	1.053	0.950
ecf_2R216	1.051	0.951
ecf_7R707	1.051	0.951
poor	1.050	0.952
ecf_7R724	1.049	0.954
ecf_3R324	1.048	0.954
ecf_3R326	1.046	0.956
ecf_5R533	1.046	0.956
ecf_3R309	1.045	0.957

term	vif	tolerance
ecf_3R311	1.044	0.958
ecf_5R509	1.043	0.959
ecf_6R601	1.041	0.960
good	1.041	0.961
ecf_3R308	1.040	0.962
ecf_6R612	1.039	0.963
ecf_5R534	1.037	0.965
ecf_1R133	1.032	0.969
ecf_5R507	1.032	0.969
ecf_6R608	1.031	0.970
ecf_2R218	1.029	0.971
ecf_5R531	1.029	0.972
ecf_1R131	1.026	0.975
ecf_6R606	1.024	0.976
ecf_1R107	1.023	0.978
ecf_4R416	1.022	0.979
ecf_7R715	1.020	0.980
ecf_7R722	1.020	0.980
ecf_5R523	1.019	0.981
ecf_7R720	1.019	0.982
ecf_6R614	1.019	0.982
ecf_3R301	1.019	0.982
ecf_3R318	1.018	0.983
ecf_4R418	1.017	0.983
ecf_5R510	1.017	0.983
ecf_2R226	1.017	0.984
ecf_5R505	1.016	0.984
very_poor_unsound	1.016	0.984
ecf_5R521	1.015	0.985
ecf_6R618	1.015	0.985
ecf_2R219	1.014	0.987
ecf_5R539	1.014	0.987
ecf_2R228	1.014	0.987
ecf_2R229	1.013	0.987
ecf_2R217	1.013	0.987
ecf_5R517	1.012	0.988
ecf_3R305	1.012	0.988
ecf_5R538	1.008	0.992
ecf_5R508	1.008	0.992
ecf_2R211	1.008	0.992
ecf_3R317	1.008	0.992
ecf_5R530	1.008	0.992
ecf_5R519	1.005	0.995
ecf_2R230	1.004	0.996
ecf_7R714	1.002	0.998
ecf_6R609	1.002	0.998

Table 16: VIF / GVIF

term	vif	tolerance
Insblsdfint	16.639	0.060
largest	5.582	0.179
smallest	5.410	0.185
large	3.160	0.316
two_plus_baths	3.082	0.324

term	vif	tolerance
small	2.961	0.338
powder_room	2.561	0.391
yb_pre_1929	2.333	0.429
qabove_avg	2.128	0.470
ecf_7R705	2.116	0.472
qbelow_avg	2.113	0.473
ecf_2R214	2.066	0.484
fireplace	1.773	0.564
ln_landratio	1.727	0.579
yb_1946_to_1960	1.605	0.623
single_family1story	1.578	0.634
hot_water	1.543	0.648
ecf_7R704	1.474	0.679
ecf_4R410	1.470	0.680
ecf_2R201	1.455	0.687
ecf_7R719	1.453	0.688
ecf_3R319	1.453	0.688
ecf_4R406	1.419	0.705
ecf_2R215	1.393	0.718
ecf_2R212	1.388	0.721
ecf_1R104	1.384	0.723
ecf_1R109	1.373	0.728
ecf_4R405	1.359	0.736
ecf_2R203	1.348	0.742
ecf_2R205	1.336	0.749
ecf_2R221	1.330	0.752
ecf_2R222	1.315	0.760
ecf_3R303	1.311	0.763
ecf_3R307	1.307	0.765
ecf_2R213	1.304	0.767
ecf_3R323	1.300	0.769
ecf_1R117	1.291	0.775
ecf_4R408	1.290	0.775
ecf_1R105	1.288	0.776
ecf_4R417	1.286	0.777
ecf_6R620	1.280	0.781
ecf_7R711	1.278	0.782
ecf_4R413	1.278	0.782
ecf_5R504	1.268	0.789
ecf_3R320	1.253	0.798
ecf_4R411	1.250	0.800
yb_post_1961	1.245	0.803
ecf_6R611	1.239	0.807
ecf_2R220	1.238	0.808
ecf_7R708	1.236	0.809
ecf_1R108	1.236	0.809
ecf_2R223	1.233	0.811
crawl	1.232	0.812
ecf_2R225	1.231	0.813
ecf_2R224	1.227	0.815
ecf_7R709	1.221	0.819
ecf_1R121	1.216	0.822
ecf_1R128	1.212	0.825
ecf_1R116	1.202	0.832
ecf_7R710	1.200	0.833
ecf_3R310	1.200	0.834
ecf_7R718	1.197	0.835

term	vif	tolerance
ecf_4R401	1.190	0.840
ecf_1R118	1.184	0.844
ecf_1R136	1.183	0.845
ecf_4R404	1.182	0.846
ecf_3R312	1.181	0.846
ecf_6R604	1.178	0.849
ecf_5R502	1.178	0.849
ecf_4R403	1.173	0.852
ecf_1R122	1.170	0.855
ecf_1R135	1.158	0.864
wall	1.152	0.868
garage	1.151	0.869
ecf_1R126	1.148	0.871
ecf_5R501	1.145	0.873
ecf_3R322	1.143	0.875
ecf_1R110	1.143	0.875
ecf_3R315	1.142	0.876
ecf_7R717	1.142	0.876
ecf_3R325	1.142	0.876
ecf_3R313	1.141	0.876
ecf_7R701	1.140	0.877
ecf_4R422	1.138	0.879
ecf_7R713	1.136	0.880
ecf_1R119	1.136	0.880
ecf_3R306	1.133	0.883
ecf_1R129	1.126	0.888
half_duplex2story	1.124	0.890
ecf_7R723	1.122	0.891
ecf_4R421	1.122	0.892
central_air	1.113	0.898
ecf_5R503	1.113	0.899
ecf_1R112	1.112	0.899
ecf_4R409	1.109	0.902
ecf_1R123	1.102	0.908
slab	1.101	0.908
ecf_4R415	1.101	0.908
ecf_2R202	1.100	0.909
ecf_1R106	1.097	0.912
ecf_6R602	1.096	0.912
ecf_7R712	1.095	0.913
ecf_2R209	1.094	0.914
ecf_5R529	1.093	0.915
ecf_4R419	1.091	0.917
ecf_7R716	1.088	0.919
fair	1.088	0.919
ecf_7R703	1.085	0.921
ecf_2R210	1.084	0.922
ecf_5R514	1.082	0.924
ecf_6R605	1.082	0.924
ecf_4R407	1.079	0.927
ecf_4R412	1.078	0.928
ecf_3R314	1.077	0.929
ecf_1R127	1.074	0.931
ecf_6R603	1.074	0.931
ecf_1R120	1.072	0.933
ecf_1R113	1.071	0.934
ecf_5R506	1.070	0.935

term	vif	tolerance
ecf_7R706	1.069	0.935
ecf_4R402	1.069	0.935
ecf_1R137	1.067	0.937
ecf_3R321	1.065	0.939
ecf_5R532	1.064	0.940
ecf_3R304	1.063	0.940
ecf_1R102	1.063	0.941
ecf_1R114	1.062	0.941
ecf_3R316	1.059	0.944
ecf_2R208	1.057	0.946
ecf_1R124	1.057	0.946
ecf_2R216	1.054	0.949
ecf_5R528	1.053	0.949
ecf_1R125	1.053	0.950
ecf_7R707	1.052	0.951
poor	1.051	0.951
ecf_7R724	1.049	0.953
ecf_3R324	1.049	0.954
ecf_5R533	1.048	0.955
ecf_3R326	1.047	0.955
ecf_5R509	1.046	0.956
ecf_3R309	1.045	0.956
ecf_3R311	1.044	0.958
ecf_6R601	1.042	0.960
good	1.041	0.961
ecf_6R612	1.039	0.962
ecf_3R308	1.039	0.962
ecf_5R534	1.037	0.964
ecf_5R507	1.034	0.968
ecf_1R133	1.033	0.968
ecf_6R608	1.032	0.969
ecf_2R218	1.030	0.971
ecf_5R531	1.030	0.971
ecf_1R131	1.026	0.975
ecf_6R606	1.025	0.976
ecf_1R107	1.023	0.978
ecf_4R416	1.022	0.979
ecf_7R715	1.021	0.980
ecf_7R722	1.020	0.980
ecf_7R720	1.019	0.981
ecf_5R523	1.019	0.981
ecf_6R614	1.019	0.981
ecf_3R301	1.019	0.982
ecf_3R318	1.018	0.982
ecf_5R505	1.018	0.983
ecf_4R418	1.017	0.983
ecf_5R510	1.017	0.983
very_poor_unsound	1.016	0.984
ecf_5R521	1.016	0.985
ecf_6R618	1.015	0.985
ecf_2R219	1.014	0.986
ecf_5R539	1.014	0.986
ecf_2R229	1.014	0.986
ecf_2R226	1.014	0.986
ecf_2R228	1.014	0.986
ecf_5R535	1.014	0.987
ecf_3R305	1.013	0.988

term	vif	tolerance
ecf_5R517	1.012	0.988
ecf_5R508	1.009	0.991
ecf_5R538	1.009	0.992
ecf_2R211	1.008	0.992
ecf_3R317	1.008	0.992
ecf_5R530	1.008	0.992
ecf_5R519	1.005	0.995
ecf_2R230	1.004	0.996
ecf_7R714	1.003	0.997
ecf_6R609	1.003	0.997

